

PumpRupee

Pricing pump energy waste in rupees

A clamp-on audit kit that tells you if a VFD will pay back for THAT pump.

No flow meter

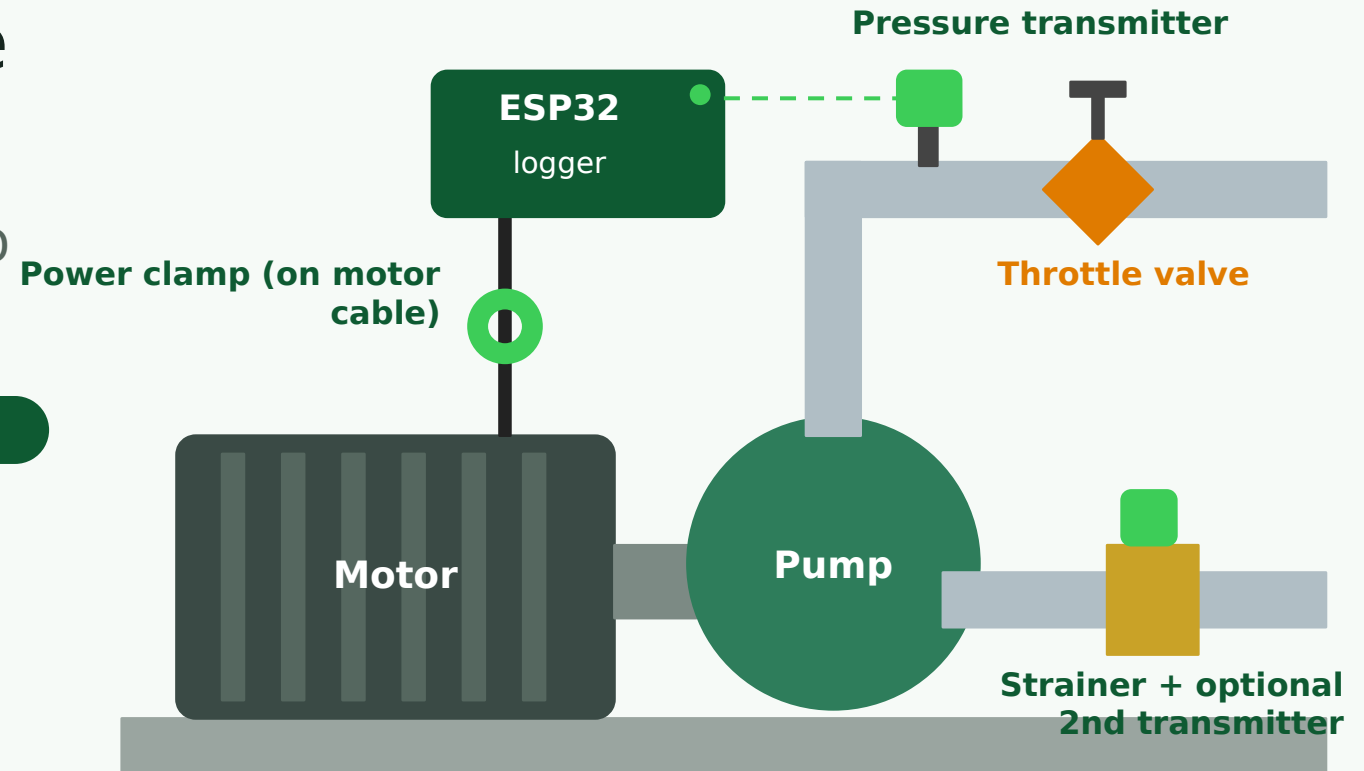
No shutdown

₹ waste by cause

Schneider Electric Yuva Yodha Energy Tech Hackathon 2026

Track: Smart Manufacturing
Electrify · Automate · Digitalise

Team ANS_4X



All results are simulated (Grundfos NB 65-160/157, 11 kW). Prices are indicative online prices, not quotes.

Problem: Invisible Waste

Oversized pumps run throttled. The extra energy is burnt across a valve, every hour.



The owner is blind to it.

Flow is hard and costly to measure. An energy-audit practitioner reports that 30-40% of ultrasonic flow readings go wrong. Head (pressure) and electrical power are easy to measure.

Most common cause

Wrong sizing leads to throttling
(practitioner report)

83-93%

of simulated waste is the valve (72-87% under stress)

8-45%

VFD saving across simulated cases; 5-40% reported in practice

Simulated figures: Grundfos NB 65-160/157, 11 kW. The 83-93% valve share is measured against a speed-matched, healthy, clean ideal.

Why Today's Options Rarely Reach MSMEs

Each option is valuable. None is a low-cost entry point that ends in a ₹ decision.

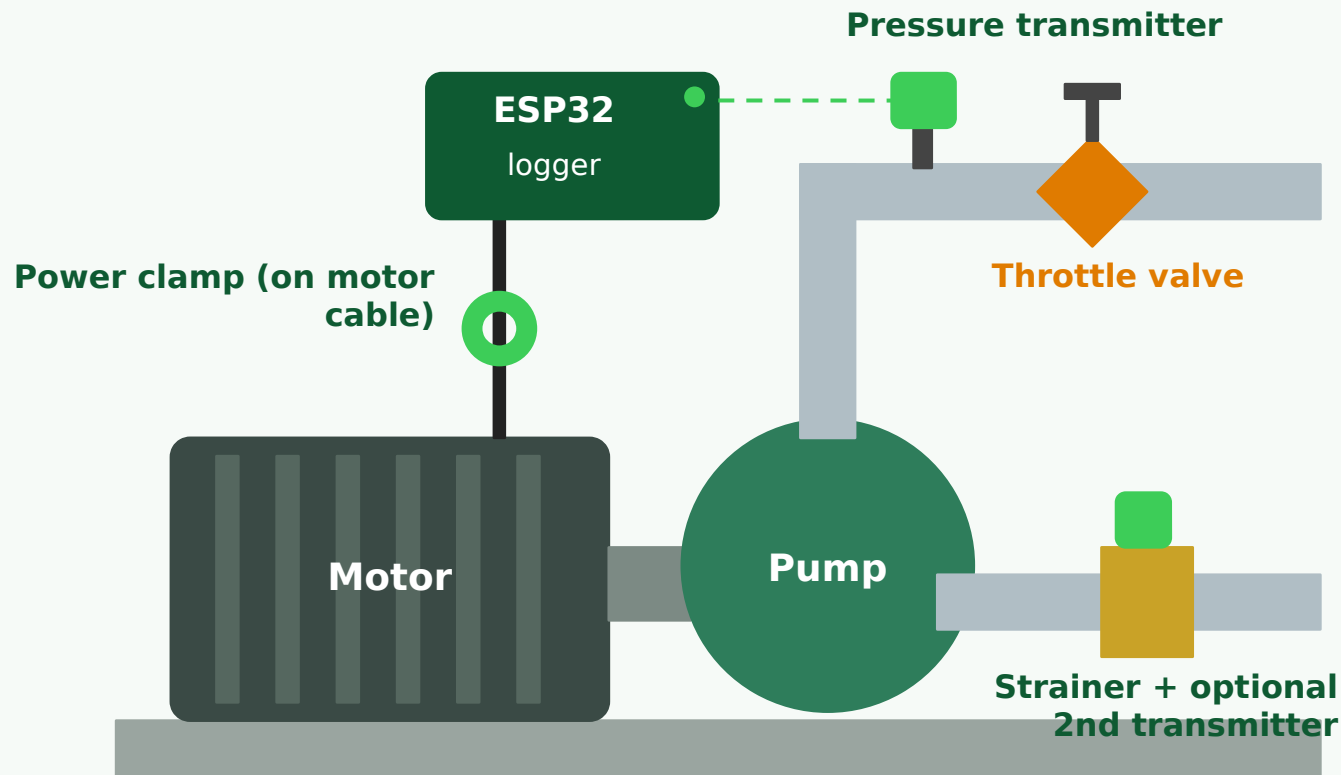
	Flow-meter retrofit	One-time manual audit	Full monitoring platform	PumpRupee
Upfront cost, small plant	Meter + install	Fee per visit	Typically higher	₹6,100-13,300
No plant shutdown	—	✓	—	✓
Answer in ₹	✗	✓	—	✓
Waste split by cause	✗	✓	—	✓
Pump-specific VFD verdict	✗	✓	—	✓
Verifies saving afterwards	—	—	✓	✓
Suited to small MSMEs	—	✗	—	✓

✓ yes ~ partly / varies by product ✗ no. Our general assessment, not a product test. PumpRupee's ticks are design targets (simulated, not field-proven).

Full platforms are the natural NEXT step after PumpRupee, not a rival.

Our Solution: A ₹ Audit Kit

A clamp-on kit turns power + pressure readings into “₹ wasted per year, and why”.



₹ waste per year

Pump energy waste priced in rupees



Waste by cause

Valve, impeller wear, clogged strainer



Cheapest fix first + setpoint

Lowest safe VFD setpoint



VFD payback verdict

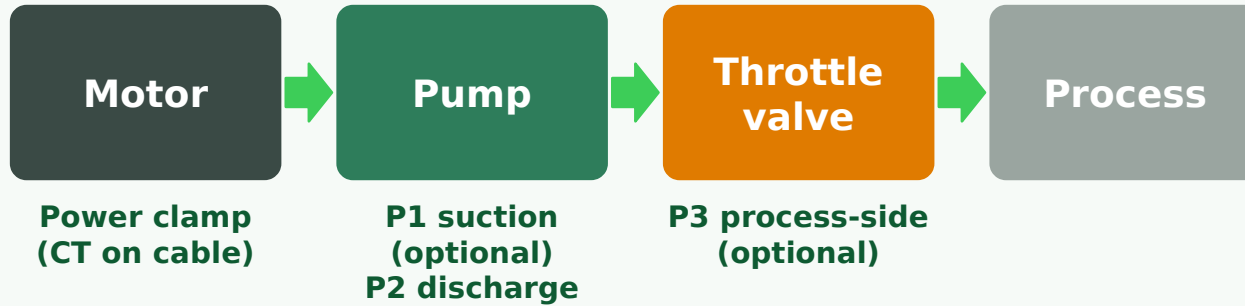
For THIS pump, before capex

VFD = control speed instead of strangling the pipe.

Audits and recommends. Does not control the pump.

How It Works: Install & Pump Curve

Clamp on, log for a week or two, and let the maths find the waste.



Minimum kit: power clamp + discharge pressure (P2).

Add P3 (process-side) when a VFD purchase is on the table: it measures the system curve and reveals a clogged strainer.

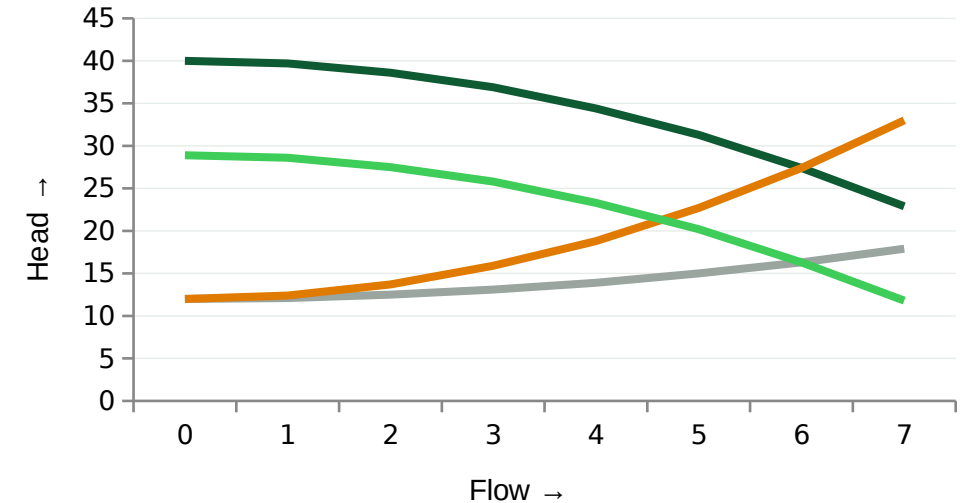
P2 + power Waste priced in ₹ (basic)

+ P3 (about ₹3,068) Error 10.5 → 1.4 pts (stress, simulated)

+ P1 / strainer ΔP Clog made visible

System curve = what the pipe network demands. Pump curve = what the pump can deliver at a given speed.

Throttled point vs VFD point (schematic)

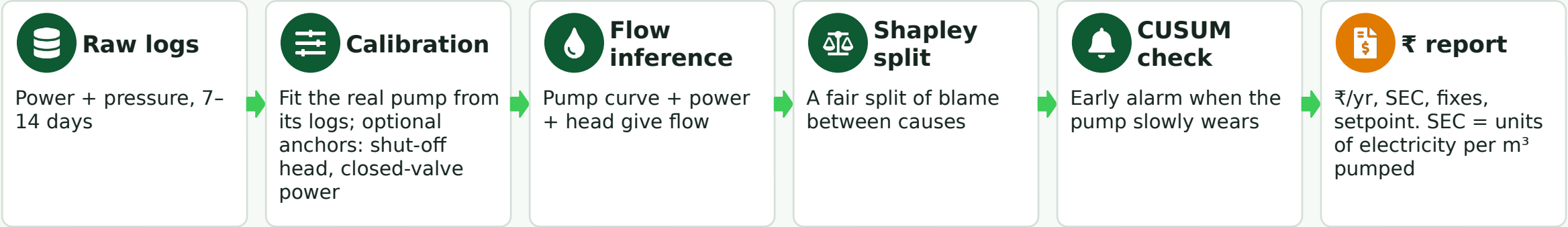


Same flow, two ways to get it.

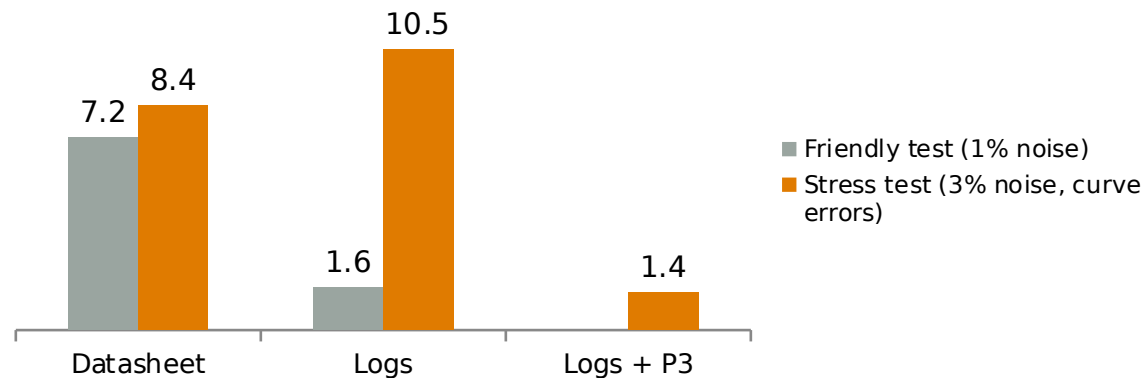
Throttled: pump pushes at full head, the valve burns the gap. VFD: the pump slows to just what the pipe needs. The gap is the wasted head, and it is priced in ₹.

Analytics & Honest Accuracy

Six plain steps from raw logs to a ₹ report, with the uncertainty stated.

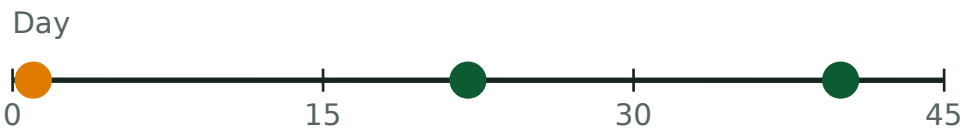


Forecast error, percentage points (simulated)



The system curve, not the pump curve, is the main error source. A $\pm 30\%$ flow reading is worse than none (16.8 vs 8.3). **About ± 3 points only with a measured system curve; otherwise about ± 10 .**

Wear-drift check (CUSUM), simulated

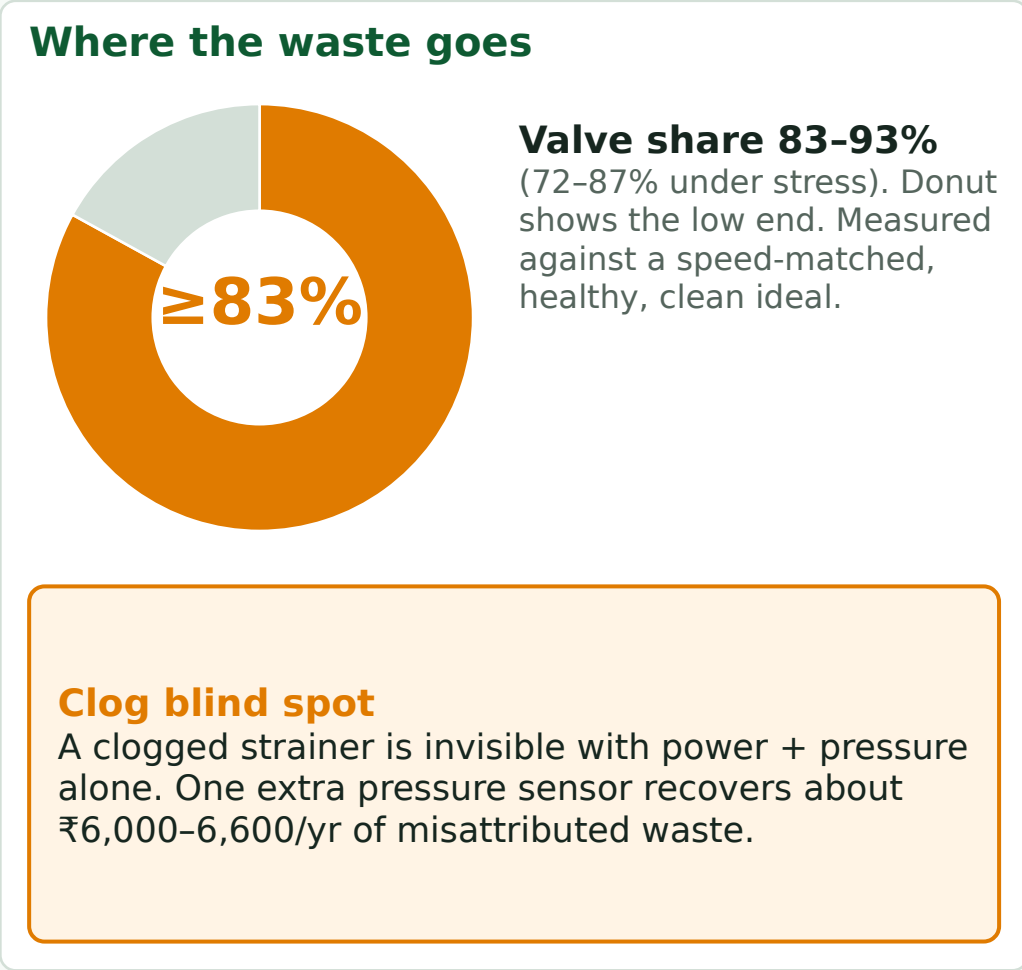
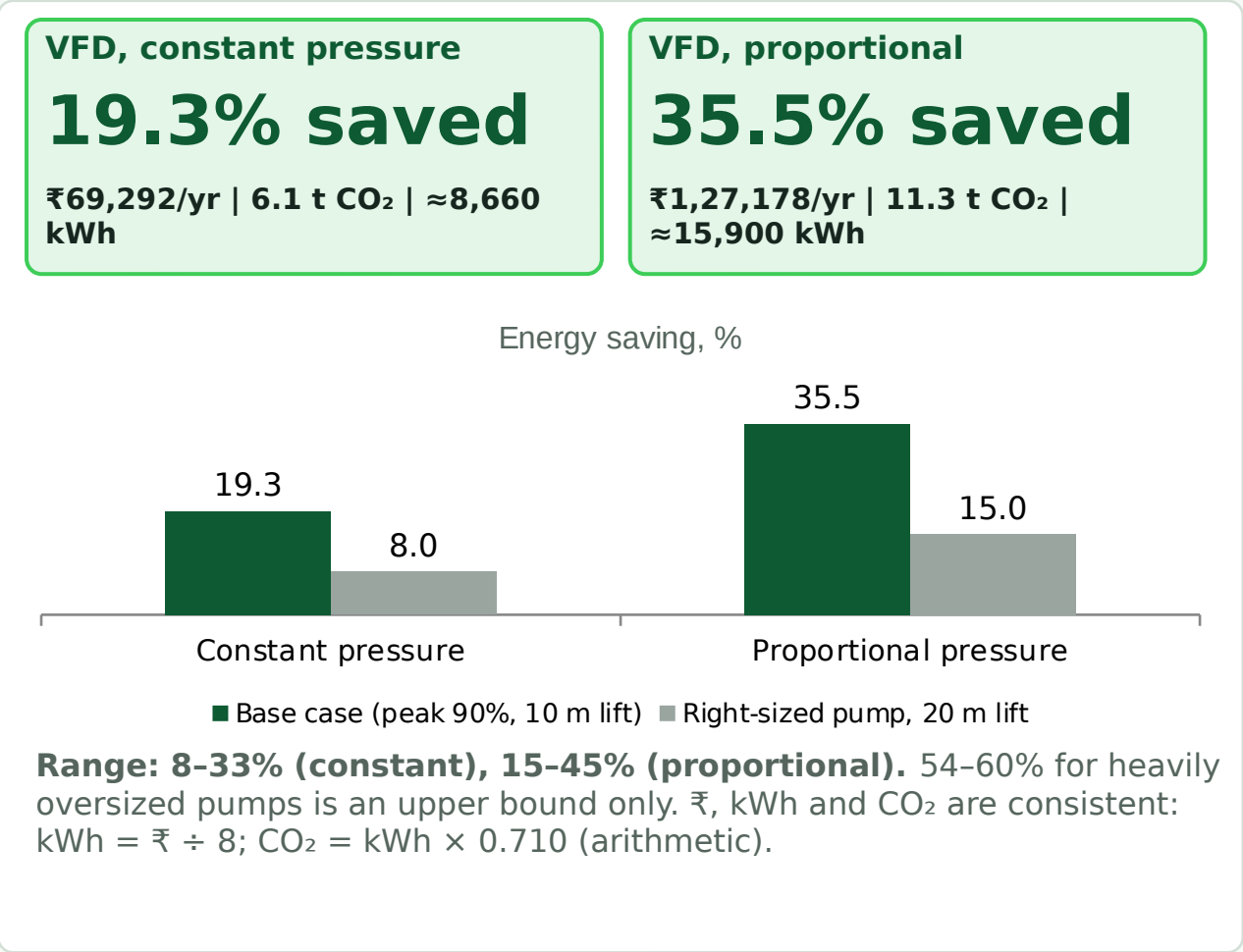


Wear caught at about 2-4%. 1 false alarm in 3 $\times$ 365 simulated days.

Results (Simulated)

SIMULATED - Grundfos NB 65-160/157, 11 kW, IE3

Assumed: 10 m static lift | 5% head margin | 16 h/day | 300 days/yr | ₹8/kWh | VFD efficiency 0.97 | 0.710 kg CO₂/kWh. Peak demand 90% of rated flow.



Cost & VFD Verdict

INDICATIVE ONLINE PRICES, NOT QUOTES

Bill of materials (preliminary)

ESP32 dev board ₹399-449

Power clamp, SCT-013 100 A ₹349 + GST to ₹729

Voltage sensor ZMPT101B ₹92-289

Pressure transmitter, basic ₹2,100-3,068

Enclosure, supply, 4-20 mA, wiring ₹2,000 (assumed)

Basic single-phase ≈ ₹6,100

3-phase + 2nd transmitter ≈ ₹10,200

3-phase + process-side (P3) ≈ ₹13,300

With ₹12,000 transmitters ≈ ₹40,100

Hobby-grade clamps; not a certified meter. Volume cost: not estimated (no quotes).

Value per pump

Cost per pump audited

Kit ÷ N pumps. N = 10: ₹610-1,330

Audit time per pump

Install (assumed ½ day) + 7-14 days logging + report

Kit throughput

About 25-50 pumps/yr per kit at 7-14 days each (arithmetic, excludes travel)

Kit payback

₹10,200 ≈ 1.8 months of the base-case saving of ₹69,292/yr, if the audit leads to that VFD (illustrative)

VFD verdict (simulated)

Oversized pump

4-19 months

GO: buy the VFD

Right-sized, high lift

2-4+ years

WAIT: fix cheaper things first

VFD ₹29,000-94,000 + about ₹15,000 install (assumed): ₹50,200-1,09,400 in total.

A ~₹10k kit can stop a wrong ₹50k-1.1 lakh VFD buy, or unlock ₹69,292-1,27,178/yr (simulated)

Ecosystem Fit: Audit → Drive → Verify

PumpRuppee is the low-cost front door to VFD and energy-management adoption.



Electrify

Makes motor electrical energy visible with a clamp-on power measurement and points owners to efficient drives.



Automate

Shows which pumps justify a drive and the lowest safe setpoint. The drive does the controlling; PumpRuppee never does.



Digitalise

Logs become a ₹ business case and a before/after proof. The owner keeps the data and can use any drive.

1

PumpRuppee audit

Clamp-on, 7-14 days

2

₹ business case

Waste by cause + VFD verdict

3

VFD installed

e.g. Altivar Process

4

PumpRuppee re-measures

Same kit, after the fix

5

Verified saving

Feeds energy management, e.g. EcoStruxure

Why this helps drive and EMS adoption

MSMEs that skip audits today get a ₹ case they understand, a pump-specific verdict before capex, and proof after.

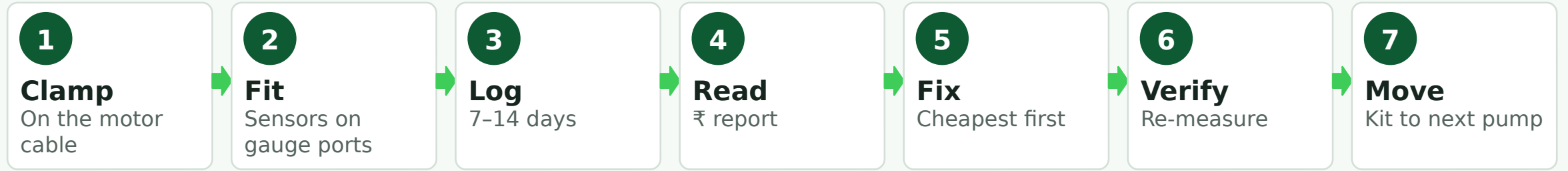
Smart Manufacturing fit

Energy efficiency · productivity · digitalisation · sustainability

Concept only: no drive or platform integration built.

Ease of Use & Safety

Clamp → Fit → Log → Read ₹ report → Fix → Verify.



The owner or technician does NOT need

- ✗ A flow meter
- ✗ Pipe cutting
- ✗ Production shutdown
- ✗ Hydraulic expertise
- ✗ Reading graphs

Safety & reliability by design

- ✓ Non-invasive power clamp
- ✓ Pressure sensors on existing gauge ports
- ✓ Rated enclosure; installed by a plant electrician
- ✓ Kit only recommends, never controls the pump
- ✓ Owner keeps the data

Closed-valve power / shut-off head anchors are optional, brief, and within the pump maker's limits.

Impact, Scalability & Roadmap

IMPACT = SIMULATED, per 11 kW pump

₹69k-1.27 lakh
per year

≈8,660-15,900
kWh per year

6.1-11.3
t CO₂ per year

Scale-up illustration, not a projection: N pumps × per-pump saving. N = 10: ₹6,92,920-12,71,780/yr and 61-113 t CO₂/yr.

Scalability

Single pump

Plant (shared kit)

Cluster / ESCO / channel partner

Bundled with VFD & EMS offers

Applications: MSME factories, water utilities, textile/dyeing, food processing, commercial HVAC chilled-water pumps. Agricultural pumping: future scope only.

Roadmap

1. Simulation
2. Bench test, real pump + sensors
3. Pilot, 1-2 plants, measured system curve
4. Validate forecast error
5. Multi-pump / multi-plant rollout

DONE

NEXT

PLANNED

PLANNED

PLANNED

Validation status

SIMULATED / DESIGNED

Savings, waste split, calibration, drift detection, cost estimates

TO BE TESTED

Flow inference vs a reference meter; throttled-valve and blocked-strainer cases; wear drift; pilots; before/after VFD saving

PumpRupee on One Page

Cheap kit. Clever inference. Answers in rupees. Honest about limits.

